# Affective Computing-Enhanced RAG for Cancer Patient Engagement: An Emotion-Aware Support System for NGOs

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***Abstract*—Cancer-support systems must provide reliable information while remaining sensitive to the emotional state of patients and caregivers. Conventional retrieval-augmented generation (RAG) systems can ground answers in trusted documents, but they generally treat emotionally different user messages in the same way. This paper presents CanSupport AI, an affective-computing-enhanced RAG framework designed for cancer patient engagement in an NGO setting. The proposed system combines cognitive query understanding, multimodal affective analysis, grounded information retrieval, emotion-aware response adaptation, and human escalation. The implemented affective module accepts written and spoken patient input and uses RoBERTa for linguistic emotion analysis and HuBERT for acoustic emotion analysis. The two streams are mapped into six higher-level affective dimensions and combined through weighted late fusion. The current implementation also incorporates linguistic and acoustic calibration features to produce an interpretable affective profile. The broader CanSupport architecture uses the resulting affective context together with retrieved Sanjeevani knowledge to guide an LLM response, while a safety layer is intended to route high-distress cases to human counsellors. The paper describes the system architecture, affective module, retrieval and response-generation design, safety model, and an experimental methodology for evaluating unimodal versus multimodal emotion recognition, retrieval quality, emotional appropriateness, grounding, and escalation behavior. Numerical performance claims are intentionally reserved for the completed evaluation stage.**

***Index Terms*—affective computing, cancer patient support, multimodal emotion recognition, RoBERTa, HuBERT, retrieval-augmented generation, RAG, emotional adaptation, human escalation, healthcare AI.**

## I. Introduction

Cancer diagnosis and treatment can create substantial informational and emotional demands for patients and caregivers. A patient asking about a hospital, treatment-related resource, or support programme may simultaneously be experiencing fear, sadness, frustration, uncertainty, hope, or other affective states. A support system that only retrieves an informational answer can therefore be factually useful while still being poorly matched to the user's emotional context.

Retrieval-augmented generation (RAG) offers a practical mechanism for reducing unsupported generation by supplying a language model with evidence retrieved from a controlled knowledge base. In the proposed CanSupport AI setting, the knowledge base is intended to contain verified material from Sanjeevani–Life Beyond Cancer, including programme information, hospital and volunteer information, and related organisational resources. The system is designed as a support and information layer rather than as a replacement for counsellors or as a clinical diagnostic system.

Affective computing provides a complementary capability: instead of asking only what the patient wants, the system also estimates how the patient is expressing their current affective state. The literature reviewed for this project indicates that transformer-based emotion recognition, multimodal affective computing, emotionally adaptive dialogue, cancer conversational agents, and healthcare RAG have each received substantial attention, but the reviewed works do not report the complete combination targeted here: patient emotion recognition, cancer-specific grounded retrieval, emotion-conditioned response generation, source grounding, and human/NGO escalation in one support pipeline [1], [2], [3], [**?**], [19], [20], [21].

The central research question is therefore: *How can patient emotional information be incorporated into a retrieval-grounded cancer-support conversational system so that responses remain factually grounded while becoming more emotionally appropriate and safety-aware?*

This paper makes five intended contributions:

1) an integrated architecture connecting cognitive query understanding, multimodal affective computing, RAG, response adaptation, and human escalation;
2) a multimodal affective module combining RoBERTa-based text analysis and HuBERT-based speech analysis;
3) a six-dimensional affective representation intended to provide a compact interface between emotion recognition and downstream response generation;
4) a grounded response-generation and safety design in which emotional context complements rather than replaces retrieved evidence; and
5) an evaluation framework for measuring emotion recognition, retrieval quality, response grounding, emotional appropriateness, and safety behavior.

The remainder of the paper presents the related literature, system architecture, implemented affective module, RAG and response-generation design, experimental methodology, limitations, and planned evaluation.

## II. Related Work

### A. Affective Computing and Emotion Recognition

Affective computing research spans textual, acoustic, visual, physiological, and multimodal signals. Recent reviews emphasize the importance of dataset quality, label documentation, explainability, and robustness when affective models are moved toward real-world applications [2], [3], [4], [5]. Multimodal emotion recognition is particularly relevant because affect may be expressed differently across modalities; however, the reviewed multimodal literature is often centered on physiological or visual signals rather than patient-support dialogue [6], [7].

### B. Text Emotion Recognition and Large Language Models

Transformer models have been applied to multi-label emotion classification, including crisis-oriented text, while recent work also highlights the subjectivity of emotion annotations and the need to address implicit emotional expression and dataset limitations [1], [8], [9], [10]. LLM-based affective-language research further suggests that language models can represent and reason about emotional information, but healthcare deployment requires explicit attention to safety, transparency, and appropriate escalation [11], [12], [13].

### C. Emotion-Aware Dialogue

Emotion-aware dialogue systems have investigated sentiment and arousal conditioning, empathetic response generation, response-act control, and emotionally informed screening or support conversations [14], [15], [16], [17], [18]. These studies support the principle that emotional information can influence response strategy rather than merely serve as a classification endpoint.

### D. Cancer Conversational Support and Healthcare RAG

Cancer-specific conversational-agent research demonstrates the potential of conversational systems for patient support, while recent cancer chatbot work explores RAG and different language models for information provision [19], [20]. The SOPHIE work is especially relevant to emotional cancer dialogue because it evaluates emotional appropriateness in a virtual cancer-patient setting, although its purpose is physician communication training rather than direct patient support [21]. Healthcare RAG systems, including BioBERT-based retrieval pipelines and behaviour-change chatbots, provide evidence for grounding generated responses in retrieved information [22], [23].

### E. Research Gap

The literature reviewed for CanSupport AI covers most individual components but largely treats them separately. The integration gap addressed by this work is the combination of multimodal patient affect, cancer-specific grounded retrieval, emotion-conditioned response generation, and human/NGO escalation. The claim is intentionally limited to the reviewed literature rather than asserting that no related system exists anywhere in the field.

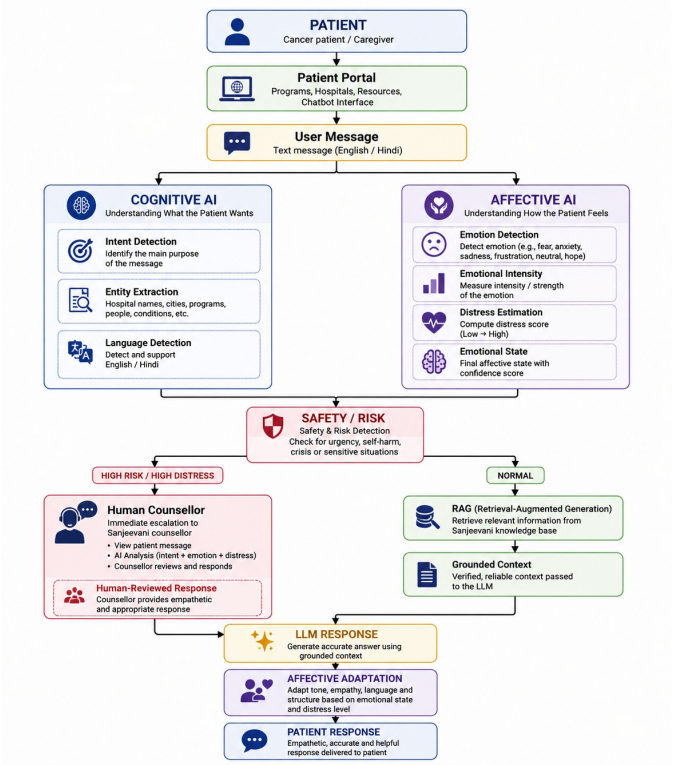



Fig. 1. Overall CanSupport AI system architecture. The diagram represents the intended end-to-end patient-support pipeline, including cognitive understanding, affective analysis, safety/risk handling, grounded retrieval, LLM response generation, and affective adaptation.

## III. CanSupport AI Framework

### A. System Overview

CanSupport AI is designed as a patient-engagement and triage layer for an NGO. A patient or caregiver interacts through a portal and submits a message. The system performs cognitive query understanding and affective analysis in parallel. Cognitive processing identifies intent, entities, and language. The affective branch estimates emotional signals, intensity, and an overall affective state. A safety/risk layer then determines whether the conversation should continue through the normal RAG path or be prioritised for human review. The intended normal path retrieves verified information, passes grounded context to the LLM, and applies emotion-aware response adaptation before returning the patient response.

The architecture follows the project requirements in which affective analysis updates an emotional-support signal, retrieval uses the patient query together with contextual information, and high-distress or low-confidence cases may be routed to a counsellor [24].

### B. Multimodal Affective Computing Module

The current affective implementation is the most concrete experimental component of the system. It accepts a written statement, a voice recording, or both simultaneously. The text branch uses a locally stored RoBERTa sequence-classification model configured for 28 outputs corresponding to the GoEmotions label space. These outputs are aggregated into six

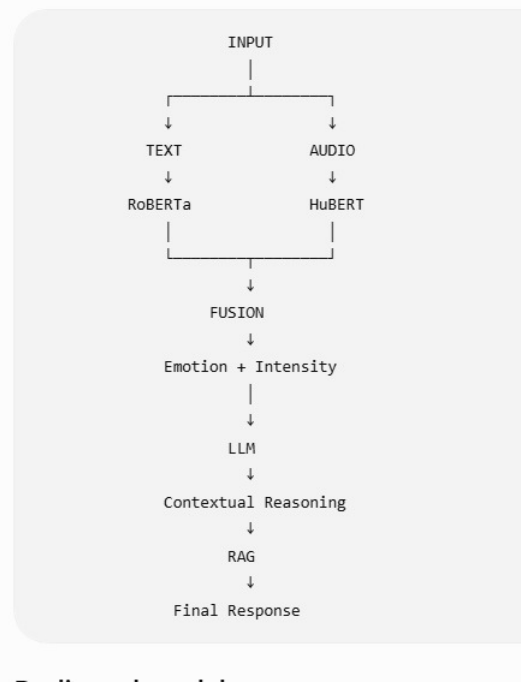



Fig. 2. Affective module pipeline used in the current implementation: written input is processed by RoBERTa, speech input by HuBERT, and the two affective streams are combined before downstream contextual reasoning.

higher-level dimensions: Happiness & Hope, Sadness & Grief, Frustration & Anger, Anxiety & Fear, Shock & Surprise, and Aversion & Disgust.

The speech branch uses the `facebook/hubert-base-ls960` backbone with a custom classifier. The implementation constructs an eight-class output corresponding to the acoustic emotion categories used in the audio pipeline and maps these outputs into the same six higher-level affective dimensions. Audio is processed at 16 kHz, with a five-second target segment used for the HuBERT input. The implementation additionally computes RMS energy and zero-crossing rate (ZCR) as acoustic calibration features.

### C. Textual Affective Representation

Let $p_g$ denote the probability assigned by the RoBERTa classifier to a GoEmotions label $g$. For each higher-level emotion $e$, the system aggregates the probabilities of the constituent labels:

$$T_e = \sum_{g \in G_e} p_g, \tag{1}$$

where $G_e$ is the label subset associated with emotion dimension $e$. For example, Happiness & Hope aggregates labels including joy, optimism, relief, gratitude, love, and related positive-affect labels. The implementation also applies a semantic lexical calibration stage using emotion-associated word lists. This calibration is part of the current prototype and should therefore be treated as an implementation heuristic rather than as a separately validated psychological measure.

### D. Acoustic Affective Representation

The HuBERT output is converted to a six-dimensional affective representation using a mapping from the eight audio classes. Acoustic calibration then modifies the affective scores according to vocal energy and zero-crossing rate. In the implementation, higher ZCR is used as a proxy for vocal tension, while low RMS energy contributes to a quieter or blunted affect interpretation. A temperature-scaled softmax is finally used to obtain normalized affective scores. These heuristics are intended to make the prototype responsive to acoustic variation; their validity requires empirical evaluation.

### E. Multimodal Fusion

When both modalities are available, the current system performs weighted late fusion. For affective dimension $e$,

$$F_e = 0.55T_e + 0.45A_e, \tag{2}$$

where $T_e$ and $A_e$ are the normalized text and audio affective scores. The fused vector is subsequently normalized across the six dimensions. When only one modality is present, its affective distribution is passed through directly.

Although the prototype code contains a comment describing the fusion as "entropy-weighted dynamic Bayesian fusion," the implemented operation is a fixed weighted linear late-fusion rule. The paper therefore describes the method according to the actual computation rather than the code comment. This distinction is important for reproducibility.

### F. Affective State and Response Adaptation

The fused six-dimensional vector provides a compact affective context for downstream processing. The intended response policy does not treat the detected emotion as a clinical diagnosis. Instead, it uses the affective state to modify communication strategy. For example, anxiety may favour validating and reassuring language without false reassurance; confusion may favour a simpler stepwise explanation; and a neutral informational query may favour a direct factual response. The project requirements explicitly define these outputs as support/prioritisation signals rather than psychiatric or medical diagnoses [25], [26].

## IV. Retrieval-Augmented Response Generation

### A. Knowledge Base and Hybrid Retrieval

The proposed RAG component is intended to retrieve information from a verified Sanjeevani knowledge base containing organisational and programme material. The project design specifies hybrid retrieval using BM25 and vector search. BM25 provides lexical matching for named entities and exact terms, while vector retrieval is intended to capture semantic similarity. Their combination is expected to reduce failure modes in which either exact terminology or semantic paraphrase dominates retrieval.

For a query $q$, the retrieval stage can be represented abstractly as

$$C = \mathrm{Rank}\left(C_{BM25}(q) \cup C_{vec}(q)\right), \tag{3}$$

where $C$ is the final grounded context supplied to the language model. The exact embedding model, vector database, chunking strategy, top-$k$ values, and any reranker remain experimental configuration items and are not claimed here as completed implementation details.

### B. Emotion-Conditioned Generation

The intended response generator receives three classes of information: the patient's query, retrieved evidence, and affective context. Conceptually,

$$y = \mathrm{LLM}(q, C, F), \tag{4}$$

where $F$ represents the six-dimensional affective state. The role of $F$ is to control response style, empathy, clarity, and support strategy while $C$ constrains factual content. This separation is central to the design: emotion should influence *how* the system responds without becoming a substitute for evidence.

### C. Grounding and Traceability

The SRS requires generated answers to be traceable to the knowledge-base documents used and specifies low retrieval confidence as a reason for escalation rather than unsupported guessing [27]. In a completed implementation, each response should therefore retain source identifiers and retrieval scores so that a counsellor or administrator can review the evidence behind an answer.

## V. Safety and Human Escalation

CanSupport AI is explicitly positioned as a support system rather than a clinical diagnostic or treatment system. The safety layer is intended to consider emotional distress, explicit risk language, and retrieval confidence. The project defines an experimental Emotional Distress Score (EDS) for support prioritisation. The SRS specifies four ranges: 0–30 normal conversation, 31–60 support recommended, 61–80 counsellor attention, and 81–100 priority human review [26].

The high-distress path is intended to route the conversation to a Sanjeevani counsellor rather than resolving it through AI alone. This design is consistent with the broader safety requirement that emotion outputs must never be represented as psychiatric diagnoses and that high-distress conversations should receive human review [28].

An important distinction for the final research evaluation is that an emotion-recognition confidence score and an EDS should not be treated as interchangeable. Emotion confidence concerns model prediction, whereas EDS is an experimental conversation-level support-prioritisation measure.

## VI. Experimental Methodology

The first draft deliberately separates implemented functionality from results that have not yet been measured. The following evaluation plan will be used to populate the results section after controlled experiments are completed.

### A. Emotion Recognition Evaluation

The affective module should be evaluated separately for text, audio, and multimodal inputs. Recommended metrics are macro-F1, micro-F1, precision, recall, and per-class F1. If the underlying text task is treated as multi-label in the training/evaluation dataset, Hamming loss and label-wise metrics should also be reported. Confusion matrices should be included for the audio classifier and for any final six-dimensional classification formulation.

The key comparison is:

1) text-only RoBERTa;
2) audio-only HuBERT;
3) multimodal weighted fusion; and
4) ablated fusion variants, if feasible.

### B. Multimodal Ablation

The fixed 0.55/0.45 fusion rule should not be assumed to be optimal. An ablation can compare text-only, audio-only, equal fusion, text-dominant fusion, audio-dominant fusion, and the proposed 0.55/0.45 configuration. If sufficient validation data are available, the fusion weights can be selected on a validation set and frozen before test evaluation.

### C. Retrieval Evaluation

For the RAG component, BM25-only, vector-only, and hybrid retrieval should be compared. Recommended metrics are Recall@k, Precision@k, Mean Reciprocal Rank (MRR), and normalized Discounted Cumulative Gain (nDCG). The evaluation set should contain representative patient-support questions whose relevant source passages are manually identified.

### D. Response Evaluation

Response quality should be evaluated across at least five dimensions: factual relevance, groundedness, emotional appropriateness, clarity, and safety. A useful ablation is to compare (i) an LLM without retrieval, (ii) RAG without affective conditioning, and (iii) RAG with affective conditioning. Human evaluators should be given a clear rubric and should not be asked to infer clinical diagnoses from the model output.

### E. Safety Evaluation

Safety experiments should include high-distress and ambiguous cases, retrieval failures, and emotionally incongruent cases in which text and speech indicate different affective signals. The desired behaviour is not to diagnose the patient, but to avoid unsupported reassurance, preserve grounding, and route cases requiring human attention appropriately.

## VII. Current Implementation Status

Table I summarizes the current state conservatively.

TABLE I
CURRENT IMPLEMENTATION STATUS BASED ON THE AVAILABLE PROTOTYPE AND PROJECT DOCUMENTS.

| Component | Status |
|---|---|
| | Text affective stream |
| Implemented prototype using local RoBERTa sequence classification and 28-label GoEmotions mapping. | |
| Audio affective stream | Implemented prototype using HuBERT with an eight-class classifier and 16-kHz preprocessing. |
| Multimodal fusion | Implemented weighted late fusion with 0.55 text and 0.45 audio weights. |
| Affective UI | Implemented interactive Gradio demonstration for text/audio/both input. |
| Overall architecture | Designed and documented in the SRS and system architecture diagram. |
| Hybrid BM25 + vector RAG | Specified in the architecture; final implementation details/results remain to be documented. |
| LLM response integration | Specified in the architecture; model/configuration and evaluation remain to be documented. |
| Human escalation | Specified through EDS and counsellor workflow; operational evaluation remains pending. |

## VIII. DISCUSSION

The principal research opportunity in CanSupport AI is not simply adding an emotion classifier to a chatbot. The intended contribution is the use of affective information as a controlled conversational signal inside a grounded cancer-support pipeline. The architecture separates three responsibilities: affective computing estimates how the patient is expressing their current state, retrieval determines what information can be safely supported by the knowledge base, and the LLM converts that evidence into a patient-facing response whose style is adapted to the affective context.

The multimodal design is motivated by the possibility that spoken delivery and written wording can differ. A patient may use positive language while exhibiting acoustic characteristics associated with distress, or may express distress explicitly in text. The current prototype includes an alignment analysis for such cases. However, these interpretations should be treated as model-generated hypotheses rather than clinical findings. Future evaluation must test whether multimodal fusion actually improves prediction and response quality over strong unimodal baselines.

A second issue is the use of hand-designed calibration heuristics. The current implementation combines neural predictions with lexical and acoustic rules. This can improve prototype behaviour on selected scenarios, but it can also introduce bias and reduce calibration if the heuristics do not generalize. The final paper should therefore report ablation results with and without calibration and should avoid describing these heuristics as clinically validated affective measurements.

## IX. LIMITATIONS AND ETHICAL CONSIDERATIONS

Several limitations remain. First, the current paper does not yet report held-out quantitative performance for the multimodal module. Second, the six-dimensional affective representation is a project-level abstraction rather than a validated clinical taxonomy. Third, emotion labels are inherently subjective and may vary with culture, language, context, and individual expression. Fourth, the audio component is influenced by recording quality, microphone characteristics, language, accent, and speaking conditions. Fifth, the proposed EDS is an experimental support-prioritisation score and must not be interpreted as a clinical psychological assessment.

The system should also avoid overclaiming from apparent multimodal disagreement. A discrepancy between linguistic and acoustic predictions is not evidence that a patient is "masking" distress; it is only an indication that the two modelled channels differ. Similarly, recommendations shown by the prototype should be framed as supportive communication guidance and not as medical or psycho-oncological diagnosis or treatment.

Privacy is particularly important because patient messages and voice recordings can contain sensitive health information. The intended production design therefore requires secure transmission, controlled staff access, auditability, and appropriate retention/deletion policies. The SRS specifies role-based access for staff and requires anonymisation for aggregate emotional-trend reporting [29].

## X. CONCLUSION AND FUTURE WORK

This paper presented the first research draft of CanSupport AI, an affective-computing-enhanced RAG framework for cancer patient engagement in an NGO setting. The system combines cognitive query understanding, multimodal affective analysis, grounded information retrieval, emotion-aware response adaptation, and human escalation. The current affective prototype provides a concrete multimodal implementation based on RoBERTa and HuBERT, with six higher-level affective dimensions and weighted late fusion.

The next stage is empirical validation. In particular, the project should measure text, audio, and multimodal emotion-recognition performance; compare retrieval strategies; evaluate whether affective conditioning improves response appropriateness without reducing grounding; and test whether the safety layer routes high-risk cases appropriately. These results will determine which parts of the proposed architecture constitute demonstrated contributions and which should remain future-work claims.

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